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Fall 2013 – CSE355 Database Systems

Quiz #1

(10.10.2013 | Duration: 45 min)

1) [40 pts] Match the following terms to the appropriate definitions

10x4

e disjointness constraint
c weak entity
d attribute
j subtype discriminator
f cardinality constraint
g degree
a completeness constraint
h identifier
i ternary
b composite key

a) Whether an instance of a supertype must also be a member of at least one subtype

b) contains two (or more) attributes

c) Depends on the existence of another entity type

d) Property of an entity

e) Whether an instance of a supertype may simultaneously be a member of two (or more) subtypes

f) Specifies maximum and minimum number of instances

g) Number of participating entity types in relationship

h) Uniquely identifies entity instances

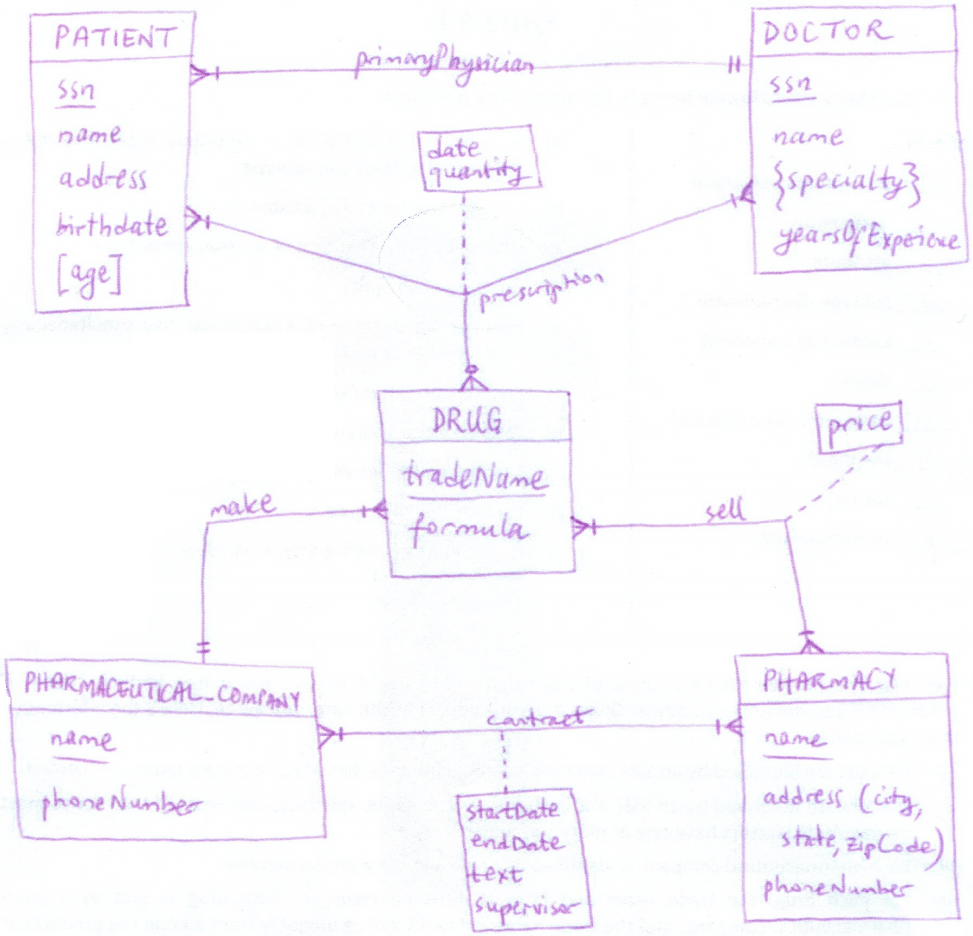
i) Relationship of degree 3

j) An attribute of the supertype whose values determine the target subtype(s)

2) [60 pts] The Prescriptions-R-X chain of pharmacies has offered to give you a free lifetime supply of medicine if you design its database. Given the rising cost of health care, you agree. Here's the information that you gather:

- Patients are identified by an SSN, and their names, addresses, birthdate, and ages must be recorded.
- Doctors are identified by an SSN. For each doctor, the name, specialty, and years of experience must be recorded. Doctors have one or more specialties.
- Each pharmaceutical company is identified by name and has a phone number.
- For each drug, the trade name and formula must be recorded. Each drug is sold by a given pharmaceutical company, and the trade name identifies a drug uniquely from among the products of that company. If a pharmaceutical company is deleted, you need not keep track of its products any longer.
- Each pharmacy has a name, address, and phone number. Address is composed of city, state, and zipcode.
- Every patient has a primary physician. Every doctor has at least one patient.
- Each pharmacy sells several drugs and has a price for each. A drug could be sold at several pharmacies, and the price could vary from one pharmacy to another.
- Doctors prescribe drugs for patients. A doctor could prescribe one or more drugs for several patients, and a patient could obtain prescriptions from several doctors. Each prescription has a date and a quantity associated with it. You can assume that, if a doctor prescribes the same drug for the same patient more than once, only the last such prescription needs to be stored.
- Pharmaceutical companies have long-term contracts with pharmacies. A pharmaceutical company can contract with several pharmacies, and a pharmacy can contract with several pharmaceutical companies. For each contract, you have to store a start date, an end date, and the text of the contract.
- Pharmacies appoint a supervisor for each contract. There must always be a supervisor for each contract, but the contract supervisor can change over the lifetime of the contract.

10 **a)** Draw an ER diagram that captures the preceding information. Identify any constraints not captured by the ER diagram.



b) How would your design change if each drug must be sold at a fixed price by all pharmacies?

5 If the drug is to be sold at a fixed price, we can add the price attribute to the Drug entity set and eliminate the price from the Sell relationship set.

c) How would your design change if the design requirements change as follows: If a doctor prescribes the same drug for the same patient more than once, several such prescriptions may have to be stored.

5 The date information can no longer be modeled as an attribute of Prescription. We have to create a new entity set called Prescription_date and make Prescription a 4-way relationship set that involves this additional entity set.